University of Florida, Department of Mathematics

Combinatorics PhD Exam, August 2025

1. Find the number of sequences of length $n \leq 26$ of letters from the alphabet, each such sequence required to be in alphabetical order with no letter repeated.

2. Let $n \geq 3$ be an odd number. Prove that the sequence

$$\binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{\frac{n-1}{2}}$$

contains an odd number of odd numbers.

3. Find a second order linear recurrence for the number of ways to properly 3-color an n-cycle. (No two adjacent vertices receive the same color.)

4. In a tournament, each player plays every other player exactly once, except in the process 3 players dropped out after playing exactly 2 games each. If 50 games were played, how many games were played just among the 3 players who dropped out? Explain your answer.

5. Suppose a convex polyhedron has 6 vertices and 12 edges. Prove that each facet is a triangle.

6. Let $A(n)$ be the number of set partitions of $[n]$ such that i and $i+1$ never occur in the same block for any $i = 1, 2, \ldots, n-1$. Show that

$$A(n) = \sum_{i=0}^{n-1} (-1)^i \binom{n-1}{i} B(n-i),$$

where $B(n)$ is the n^{th} Bell number. Alternatively, show that

$$A(n) = B(n-1).$$

7. Let P be a finite poset with a bottom element 0. Show that if x covers exactly one element of P, then

$$\mu(x, 0) = \begin{cases} -1 & \text{if } x \text{ covers } 0, \\ 0 & \text{otherwise.} \end{cases}$$

For any $n \in \mathbb{Z}$, construct a poset containing an element x with $\mu(x) = n$.

8. Prove that a simple k-regular graph with girth k has order at least $2k$.